TECHNICAL REPORT: MITIGATION OF UNPREDICTABLE PRESSURE SPIKES IN BATES

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1. Introduction

- Background:

  Experimental rocketry requires high ballistic consistency for safety and performance. The BATES (Ballistic Test and Evaluation System) grain geometry is specifically engineered to provide a neutral, stable thrust profile through a maintained burning surface area. However, practitioners often observe erratic, non-neutral pressure-time curves characterized by sudden spikes and deviations from simulation. These phenomena indicate that the propellant is not burning according to its geometric design.

- Objectives:

  This research aims to address these performance instabilities by ensuring grain structural integrity and achieving predictable combustion kinetics through the optimization of the propellant matrix and ignition systems.

2. Technical Discussion

The transition from erratic pressure spikes to predictable combustion requires managing both the physical state of the grain and the dynamics of the ignition process.

2.1 Rheological and Structural Integrity

The formation of voids and micro-cracks is the primary driver of unpredicted pressure surges.

- Surfactant Integration: Molten sorbitol exhibits high surface tension, which can hinder the wetting of KNO_3 particles and lead to air entrapment. The addition of surfactants, such as Sodium Laureth Sulfate (SLES) or Lecithin, serves to reduce this interfacial tension. This promotes a homogeneous, void-free slurry that ensures the grain burns uniformly without localized hot spots.

- Plasticization: Sorbitol inherently forms a brittle, crystalline matrix that is susceptible to thermal shock and mechanical fracture. Incorporating plasticizers like glycerol introduces flexibility into the binder, allowing the grain to resist crack propagation under operating pressures. By maintaining a crack-free structure, the propellant preserves its intended burning surface area.

2.2 Combustion Kinetics and Ignition

Controlling the burn rate involves moderating both the thermal output and the chemical initiation.

- O/F Ratio and Particle Size: Shifting the Oxidizer-to-Fuel ratio from the standard 65:35 toward a 60:40 or 58:42 composition increases the fuel-rich binder content, which acts as a thermal heat sink to moderate peak combustion temperatures. Simultaneously, using coarser KNO_3 particles prolongs the melting phase, resulting in a more controlled, gradual release of oxygen.
- Ignition Reliability: Delayed ignition contributes to non-uniform pressure rises. Implementing a two-part ignition system—consisting of a high-sensitivity primer and an internal pyrogen charge—ensures that the entire grain surface initiates combustion simultaneously, preventing the localized pressure surges associated with slow or uneven ignition.

3. Ignition System Design: Rationale and Diagnostic Criteria

The propulsion system employs a two-part ignition sequence consisting of an initiator (igniter) and a surface-applied propellant primer. This configuration is critical for achieving reliable, synchronized ignition across the grain surface and mitigating the risks associated with transient ignition delays.

3.1 Necessity of the Two-Part System

The requirement for a dual-stage system arises from the physical and chemical properties of the solid propellant grain, which often possesses high auto-ignition temperatures or geometries that challenge uniform flame propagation.

- Sensitivity Bridging: The primer serves as an intermediary layer with high sensitivity to thermal flux. The igniter provides the initial high-energy output required to trigger the primer, which in turn acts as a high-intensity catalyst to force the main propellant grain into self-sustaining combustion.
- Simultaneous Flame Propagation: Reliance on a single igniter can result in localized ignition, leading to uneven burning or combustion instabilities. The primer ensures the

flame front spreads instantaneously across the entire exposed propellant surface area, facilitating a rapid and controlled pressure rise to design levels.

- Mitigation of Hang-Fire: The system design proactively prevents "hang-fire" scenarios—where ignition is delayed beyond the specified time window—by ensuring that energy transmission is uniform and effective across the combustion chamber.

3.3 Diagnostic Identification of Delayed Ignition

Indicator	Characteristic Behaviour	Interpretation
P-t Slope	A distinct "step" or "plateau" in the pressure curve immediately following the activation signal	Indicates a lag between the initial igniter discharge and the onset of stable grain combustion.
Ignition Interval	Time duration from the electrical activation signal (t_0) to the achievement of 10% of maximum chamber pressure (P_{max}).	Exceeding the design-specified threshold for this interval constitutes a confirmed ignition delay.
Acoustic/Vibration Data	Sharp, irregular pressure spikes or "chuffing" sounds	Suggests non-uniform flame propagation or the accumulation of gases before sudden, violent ignition of the grain.

Monitoring these parameters ensures that the transition from initial ignition to steady-state interior ballistics remains within established safety and performance envelopes.

4. Conclusion

The erratic ballistic performance observed in BATES-configured KNSB motors is largely a result of structural defects and inconsistent initiation dynamics. By focusing on grain structural integrity—specifically through the use of surfactants to eliminate voids and plasticizers to prevent fractures—and refining combustion kinetics via O/F ratio adjustments and improved ignition, the motor's actual chamber pressure can be aligned with simulated design predictions

5. References

- Nakka, R. (n.d.). "KN-Sorbitol Propellant." Richard Nakka's Experimental Rocketry Web Site.
- European Conference for Aeronautics and Space Sciences (EUCASS). (2019). "Laser Ignition and Combustion Study of KNO3-Sorbitol based Solid Propellant".
- Technical Report: "Surfactants & Plasticizers for KNSB Solid Propellant" (The Nakuja Project, 2026).